SMART AND SECURE VOTING SYSTEM USING CYBERSECURITY

SEMINAR REPORT

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EXAMINER 1 **EXAMINER 2**

ABSTRACT

Election is the mechanism which is used by people to choose a leader. The citizens can cast their vote either through ballet paper or Electronic Voting Machine [EVM]. Using either way of voting has a chance to misuse the other votes. As a result, this research work proposes a fingerprint sensor and face recognition-enabled electronic voting system. This helps to conduct a safe election without rigging as fingerprint and face are used as identification features. The vote caster fingerprints will be taken and it will be extracted and saved in the database for the voter registration and authentication processes. Storing the information in the database will prevent more than one registration from a single person. At the time of voting or during voting day, the individual person need to scan his/her fingerprint and it will be compared with the fingerprint database if the fingerprint is found in the fingerprint database. By using the Voter identity number and fingerprint, numerous duplicate registrations can be stopped and this will lead to high rate of actual success during voting. Voters can cast their ballots from any part if world by using their voter's unique identifier, authentication responses given on enrollment, and a token key supplied immediately for every member via the specially designed election web module. The administrator will send the token to the linked candidate's Email address. Some extra features are added to the project when compared with the existing systems.

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1. INTRODUCTION

Electronic Voting Machines (EVMs) were introduced in Indian elections between 1998 and 2001, in a phased manner. Before this, India used paper ballots and manual counting. The paper ballot method was widely criticized because of fraudulent voting and booth capturing. EVMs have helped reduce electoral fraud and abuse, eliminate booth capturing and create more competitive and fairer elections. Voter-verified paper audit trail (VVPAT) system was tried on a pilot basis in the 2014 Indian general election. Supreme Court of India ordered the Election Commission of India to use the VVPAT paper trail system in every assembly constituency but verify only about 2% of the EVMs before certifying the final results. The Election Commission states that their machines, system checks, safeguard procedures, and election protocols are "fully tamper-proof".

The cost per EVM was ₹5,500 at the time the machines were purchased in 1989–90. The cost was estimated to be ₹10,500 per unit as per an additional order issued in 2014. Even though the initial investment was heavy, it has since been expected to save costs of production and printing of cores of ballot papers, their transportation and storage, substantial reduction in the counting staff, and the remuneration paid to them. For each national election, it is estimated that about 10,000 tons of ballot paper are saved. EVMs are easier to transport compared to ballot boxes as they are lighter,

more portable, and come with polypropylene carrying cases. Vote counting is also faster. In places where illiteracy is a factor, illiterate people find EVMs easier than the ballot paper system. Bogus voting is greatly reduced as the vote is recorded only once. The unit can store the result in its memory before it is erased manually. The battery is required only to activate the EVMs at the time of polling and counting and as soon as the polling is over, the battery can be switched off. The shelf life of Indian EVMs is estimated at 15 years.

2. PROBLEM STATEMENT

In India, every citizen is allowed to exercise the power of choosing there representative through easting votes. All voting systems currently conduct the following steps: voter identification and authentication, vote easting, vote recording, vote counting, and election results posting. The electoral process is centered on security. As a result, the development of a safe electronic voting machine is critical. Hence, our votes are casted; we sure ensure that they are not manipulated in any way. In this project we will try to solve this vulnerability by introducing a three steps verification which will help to avoid the middle man attack that is by using face recognition, fingerprint authentication and OTP verification. Only when the voter is physically present then the fingerprint of the user is verified and the unique identity of the user that is "Face Recognition" if matched with the stored database, then the user will receive OTP to the registered mobile number then he/she will have the power to east their vote. If any of the above verification is not valid then he/she will not able to exercise the voting power. It ensures the transparency and high security, so there will be no malfunctioning during the election process. The proposed voting mechanism has a number of advantages over the old approach.

3. SCOPE AND OBJECTIVE

To use fingerprint-based authentication and facial recognition for secure voting. For a more effective and transparent election process, fingerprint-based authentication and face recognition-based data exchange are used to address security concerns in electronic voting machines. The proposed strategy achieves better results and gives candidates more opportunities to vote in any place. To develop a fingerprint and face-recognition-based advanced EVM. Casted votes will be counted automatically at the end of the voting process. The result will be generated centrally with less time. To transfer voting information, we employ GSM technology. The RFID tag will be scanned and matched to the finger with the Aadhar database. The main motto of the project is to develop a system to avoid fake votes and detect whether the person is eligible to vote or not by detecting their biometric features (fingerprint, facial).

4. EXISTING SYSTEM

The existing system is not too effective. At present there are two types of voting methods, they are:

A ballot is a device used to cast votes in an election, and may be a piece of paper used in secret voting. In this the voter is given a paper which consists of all the party symbols along with representative names in it. Here, people come to the polling booth, take the ballot paper and vote by putting a stamp on the desired party symbol. Finally, the ballot paper is folded and dropped into the ballot box. At last, the votes are counted by the Election commission officers.

An EVM is a device which is used for voting. This machine consists of party symbols along with the representative's name and a button at the end for each and every party name. The voters come near the EVM machine after completion of their verification at the early level before voting. After verification the voter goes near the EVM and casts their vote by pressing the button. The above procedures are not so accurate as there may be possibility for the false/fake voting. As soon in Fig. 4.1

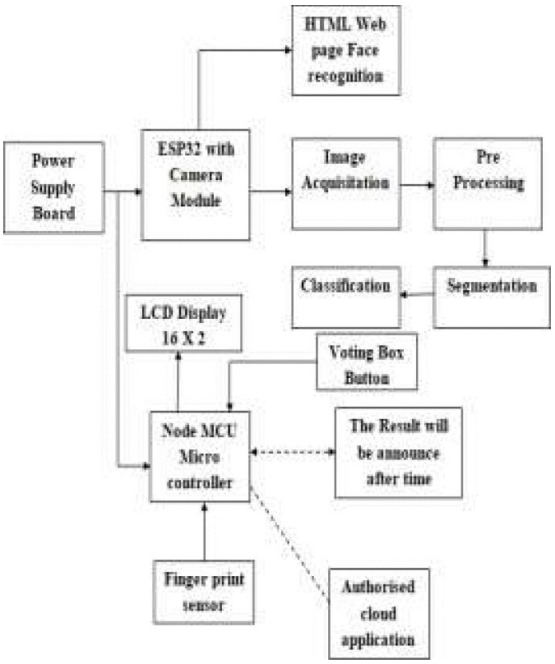


FIG 4.1 : Existing Diagram

5. LITERATURE SURVEY

TABLE 5.1 : Literature Survey Table

Sl. No.	Paper Name	Year of Publishing	Objective	Advantage	Disadvantage
1.	Electronic Voting System using Biometrics, Rasberry pie and TFT module	2019	When voter votes for a candidate, the vote count is incremented and encrypted before giving it to central database and user disable request is given to system to avoid multiple voting	This proposed " Vote from Anywhere" system enables voter to cast his vote from anywhere in India just by visiting to the voting center to use this system.	Double voting is possible here. Admin can make use of web portal to check vote count and to update and insert information of candidates.
2.	A Secured Biometric Voting System using RFID Linked with the Aadhar Database	2020	They proposed using an RFID tag instead of a voter ID. The RFID tag will be scanned and match the finger with the Aadhar database	If the finger matches with the Aadhar database, the person can cast their vote	We have to create RFID tags for each individual person, which is costly
3.	Secured E-Voting System using Two-Factor Biometric Authentication	2020	A 2-factor Biometric authentication system Face using Eigen face based recognition algorithm and	This can be used for both online voting and traditional polling booths.	We have to create Eigen face based tags for each individual person, which is costly.

			Minutiae based algorithm with a full pledged database as an input.		
4.	Smart and Secured Voting Machine using Biometrics	2020	One of the major problems in existing voting system, which is EVM, we propose a system or method which will solve the problem of the current voting system available in India	This system uses a cloud database; hence it will be more efficient than the existing system. There is additional GPS in the system, this will prevent the system from theft if the system is stolen it can be located easily and immediately through satellites	In this system, we can add any other device for better performance. This system can be further improved with any other advance technology. So that the system can be more efficient, and hence can be more secure and compact
5.	Biometric Based Secured Remote Electronic Voting System	2020	The system proposed in the present paper, shall serve with a set of innovation advantages namely, Reduce rigging and fake invalid votes, ease of carrying	The fingerprint based voting system that has been proposed in this paper is faster and more efficient	A fingerprint based remote EVM using Arduino is proposed to ensure fair and square election process.

			the machine and faster and more accurate voting process and remote voting capability	than the systems reported in the literature previously	
6.	Secured and Transparent Voting System Using Biometric And Face Recognition	2021	Biometric approach eliminates fraud and illegal behaviors, which is a major problem in the traditional voting system. The proposed methodology ensures that the counting procedure is transparent and is more accurate.	The vote count is incremented and the vote is also encrypted. The user needs to enter his Aadhar details on the touch screen available for voting	For educated people, it is easy to operate, but for the uneducated, it's not that much easier to operate
7.	ElectionBlock: An Electronic Voting System using Blockchain and Fingerprint Authentication	2021	The design and development of Election Block, a voting system that provides its own block chain, running on a centralized network of nodes, with the integration of a biometric	The purpose of the application is to enhance the current ballot based voting system used by most large scale election	Vote anonymity and security are also largely handled due to the blockchain platform.

			scanner to maintain vote integrity		
8.	Enhancement of Electronic Voting Machine performance using Fingerprint and Face recognition	2021	This paper presents, the enhancement of Electronic Voting Machine performance using Fingerprint and face recognition	This system allows only authenticated voting than the existing equipment as the person is identified based on his fingerprint and Face recognition which is unique to each individual.	To overcome this problem in this project EVM, Right now, the government of India taking a lot of effects to make 100% polling. Due to a lack of facility in the EVM, it is not possible.
9	Biometric based Smart and Secured Electronic Voting Machine	2021	This article analyzes absolute assessment about projecting a voting form devices, topic and among the majority rule procedures and biometric EVM	This can be used for projecting a voting form since it rout every one of the weaknesses of ordinary majority rule machine moreover give additional security.	This can be ised for projecting a voting form since it route everyone of the weaknesses of ordinary majority rule machine moreover give additional security.

10.	Touchless Electronic Voting Machine with an AI Facial Recognition	2021	build a smart voting system using face recognition technology that allows any voter in INDIA by going to their respective constituency to the nearest voting booth in the place of stay.	The method involves new proposed is simple and ways to use the technologie s already in existence	In the field of data-processing and image-processing to improve the simultaneous re-cognition of multiple images...
11	Smart Voting Machine using Fingerprint sensor and Face recognition	2022	In this paper, authors have eliminated rigging of voting. They have attached the GSM module with a fingerprint sensor.	If RFID key is missed then voter can't cast their vote	We have to create RFID tags for each individual person.
12	Secure Electronic Voting Machine using Biometric Authentication	2022	In this paper, the design of an Electronic Voting Machine (EVM) is proposed that will be able to mitigate the problems and to strengthen the trust of the citizens in the electoral process.	It integrates biometric information i.e. fingerprint in the EVM which eliminates the impersonati on, counterfeiti ng, and tampering issues.	Further, we can enhance the system by adding other biometric technologies such as face recognition, iris recognition, etc.

6. ARCHITECTURE

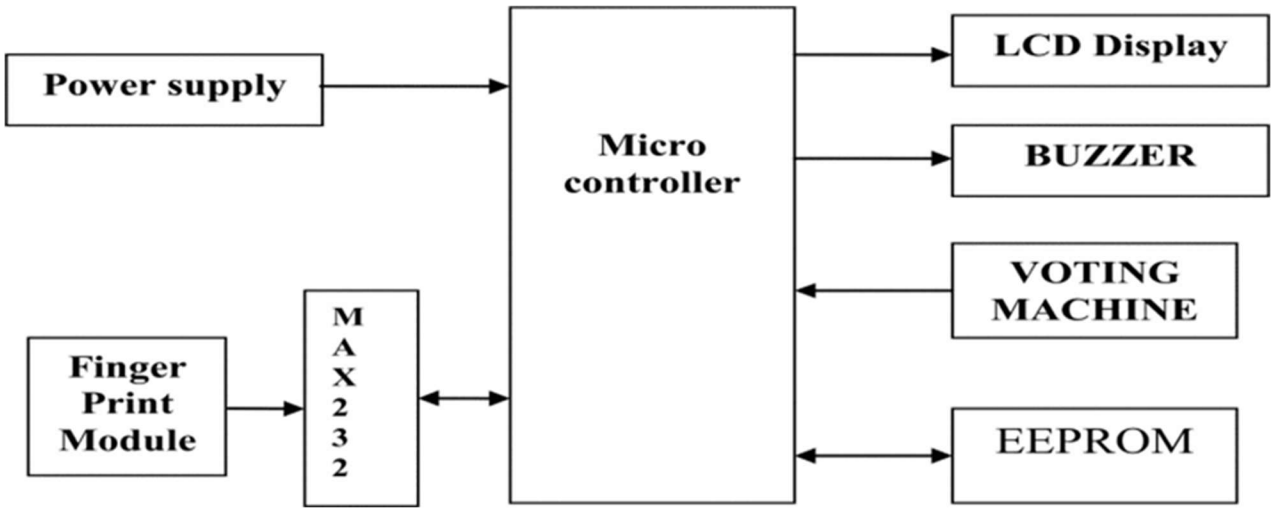


FIG 6.1 : Architecture Diagram

From the above Fig 6.1, The procedure provides a platform for certified voters to input information and fingerprint registration follows swiftly. The system spontaneously assigns a face ID or the RFID to that voter in the system. Voters register in this phase by providing their personal details and fingerprint. Some fields such as surname, first name, and sex are compulsory parameters. After enrolment, a PIN is assigned to each user by the system. All registration details are saved in the database. At electoral phase, the voter is granted access into the voting system by fingerprint verification, and if valid he/she is allowed to vote. The vote once casted is encrypted with a private key. The encoded vote is embedded into a cover photo

which is then directed into the database to be sorted. The final phase is the where votes casted are verified to satisfy they are unaltered during embedding process. The casted votes are retrieved and sorted, which is published for the electorate.

7. PROPOSED SYSTEM

In this system, the interface accepts the fingerprint data and facial data of the voters and voter's Aadhar card number. On successful authentication, the interface allows to vote and displays confirmation message. If any problem occurs, then an error message is displayed based on the type of problem. The fingerprint data and facial data is used for authentication because fingerprints and facial processing is faster, efficient and is unique to each individual. A central database is maintained which contains all the demographic and biometric data of every voter enrolled. In order to reduce the load central database is subdivided into many local databases based on the center of individual voters. The data is periodically updated and stored in volatile form so that it can be erased and retrieved only when the data is needed. The databases will be used for generating reports and result of the electoral process can be viewed by only an authorized person. To authenticate, valid national ID card number is checked in the local database. If person number is not found then person will not be allowed to take part in the voting process. The databases also count the number of votes to each party and finally generate the result. This process is shown in Fig. 7.1

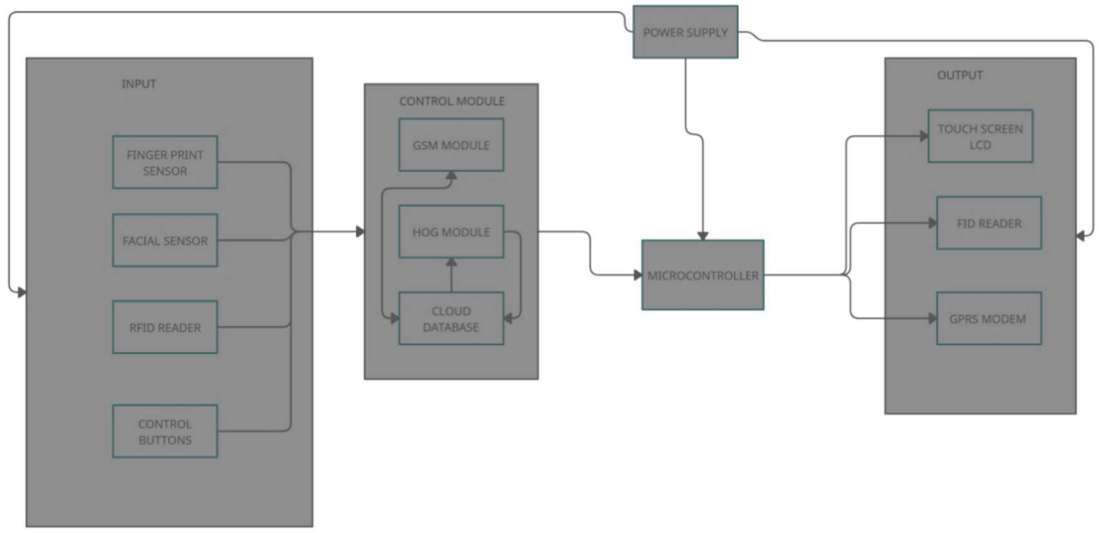


FIG 7.1 : Proposed Diagram

8. SYSTEM SPECIFICATION

Hardware:

- MICROCONTROLLER
- IMAGE SCANNER
- FINGERPRINT SCANNER
- LCD DISPLAY SCREEN

Software:

- GSM (GLOBAL SYSTEM FOR MOBILE COMMUNICATION)
- HOG(HISTOGRAM ORIENTED GRADIENT)
- RFID READER

9. MODULE DESCRIPTION

9.1 Input Unit:

9.1.1 Fingerprint Sensor:

In this section, the proposed fingerprint recognition based on Minutiae algorithm as explained below, the majority of known fingerprint extraction methods are based on ridge structure. There are mainly two steps which are widely used in fingerprint sensor,

1. Fingerprint Template formation.
2. Fingerprint matching algorithm.

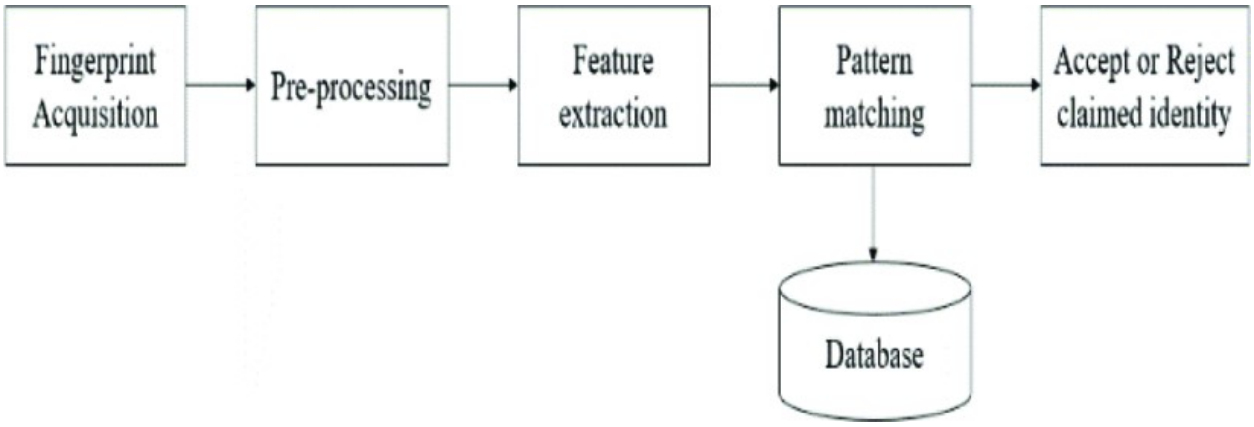


FIG 9.1 : Real time fingerprint authentication block diagram

Minutiae extraction is based on marking the points where ridges ending, ridge

bifurcation, small ridge, crossover, etc. Mainly we concentrate on the ridge termination and ridge bifurcation. The accuracy of fingerprint identification improves with this information.

Only a section of the fingerprint image is relevant in terms of providing the required information and being helpful for further processing. The ROI, or region of interest, refers to this area. The area that contains merely background information, such as ridges and furrows, is removed during this operation. A gray scale picture fingerprint is obtained from the sensor.

9.1.2 Facial Sensor:

In this EVM operation, three algorithms are used separately for different actions. Like, Face recognition, fingerprint recognition, and pushbutton identifications. These programs are implemented using open CV libraries.

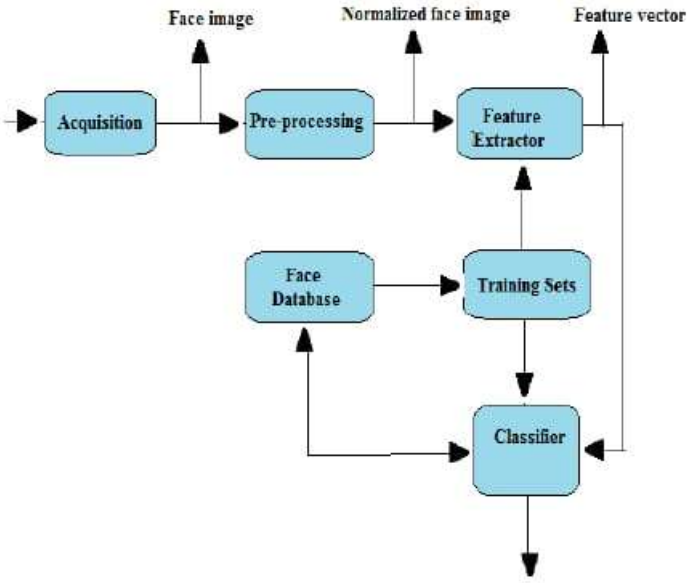


FIG 9.2 : Face recognition block diagram

The process starts with a request for the name to be entered to be stored with the ID in a text file. The face detection system starts in the first half. However, before capturing begins, the application checks for the brightness levels and will capture only if the face is well illuminated. Furthermore, after the face is detected, the position of the eyes is analysed. If the head is tilted, the application automatically corrects the orientation. These two additions were made considering the requirements for the Eigen's face. The image is then cropped and saved using the ID as a filename to be identified later. A loop runs this program until 50 viable images are collected from the person. This application made data collection efficient.

9.1.3 RFID Reader:

RFID (Radio Frequency Identification) is indicated in the figures and is designated to the automated identification and information capture technologies. This is the secure and fast approach of identifying the various objects. The Interrogator (RFID Reader) is capable of transmitting and receiving information and the transponder tags, which are attached to objects and things. RFID tags include the miniscule microchips and the antenna. The tags might be passive or active and is available in large variety of sizes and shapes. The information transfer between the tags and readers takes place wirelessly and does not necessitate any line of sign in

the devices. The Reader circuit declaims through the various tags mobile and fixed, except for the conductive components such as metals and water. It gives out low power radio waves, which are utilized to power the tags to pave way for data that have been stored in the chip. Generally, the passive tags are lighter, expensive and smaller compared to the active tags and are applicable to things in the hazardous environment. Moreover, their cost of maintenance is affordable and it is also durable. In this case, the systems utilize VLF RFID tags that are applied to the 125 kHz ranges. The tags include a unique 32-bit number that is nonprogrammable. The card is smooth, mildly flexible and smooth.

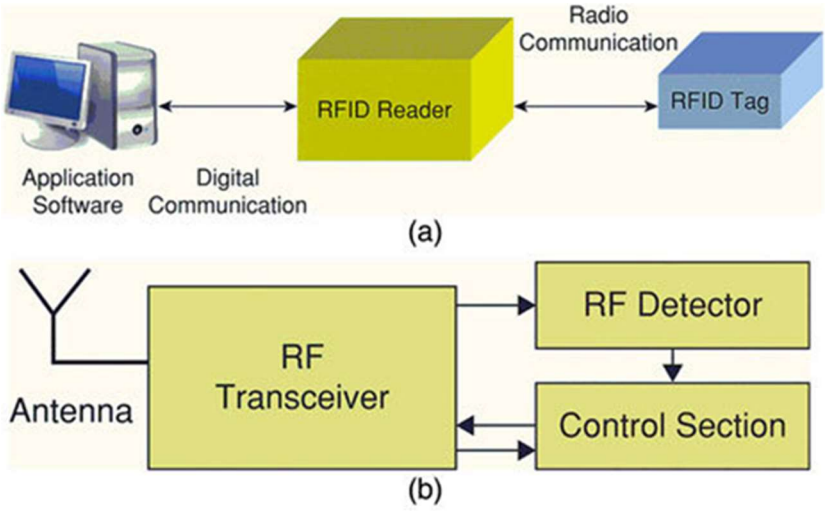


FIG 9.3 : RFID READER WORKING

9.1.4 GSM Module:

A GSM module or a GPRS module is a chip or circuit that will be used to establish communication between a mobile device or a computing machine and a GSM or GPRS system is a standard developed by the European Telecommunications Standards Institute (ETSI).It was created to describe the protocols for second-generation (2G) digital cellular networks used by mobile phones and is now the default global standard for mobile communications – with over 90% market share, operating in over 219 countries and territories.

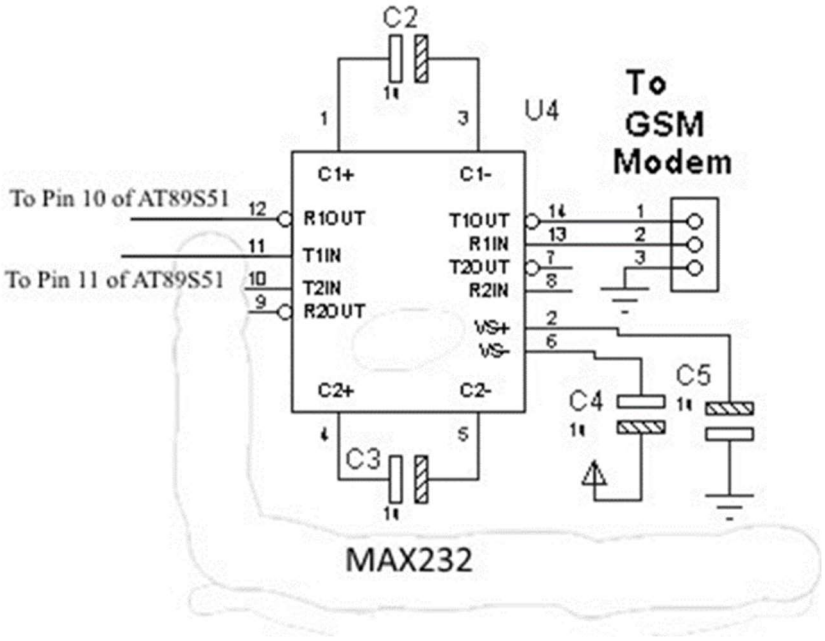


FIG 9.4 : GSM Modem

GSM modem sends received message to microcontroller. Microcontroller keeps record of all valid voters ID. AT command is used as protocol for communication

between GSM modem and microcontroller Then, microcontroller decodes message received from GSM modem.. It then compares voters ID of message with IDs in its own database. Microcontroller checks whether received voter ID is valid or not. If it is valid then it further checks whether vote is already received from this voter ID. If microcontroller finds that vote is already received then it sends message to voters Id that Vote is already received from this ID.

9.1.5 HOG Module :

Histogram of Oriented Gradients (HOG) is a feature descriptor used in image processing, mainly for object detection. A feature descriptor is a representation of an image or an image patch that simplifies the image by extracting useful information from it. The principle behind the histogram of oriented gradients descriptor is that local object appearance and shape within an image can be described by the distribution of intensity gradients or edge directions. The x and y derivatives of an image (Gradients) are useful because the magnitude of gradients is large around edges and corners due to abrupt change in intensity and we know that edges and corners pack in a lot more information about object shape than flat regions. So, the histograms of directions of gradients are used as features in this descriptor. As per Fig.4.2.6 that is how the HOG software works .

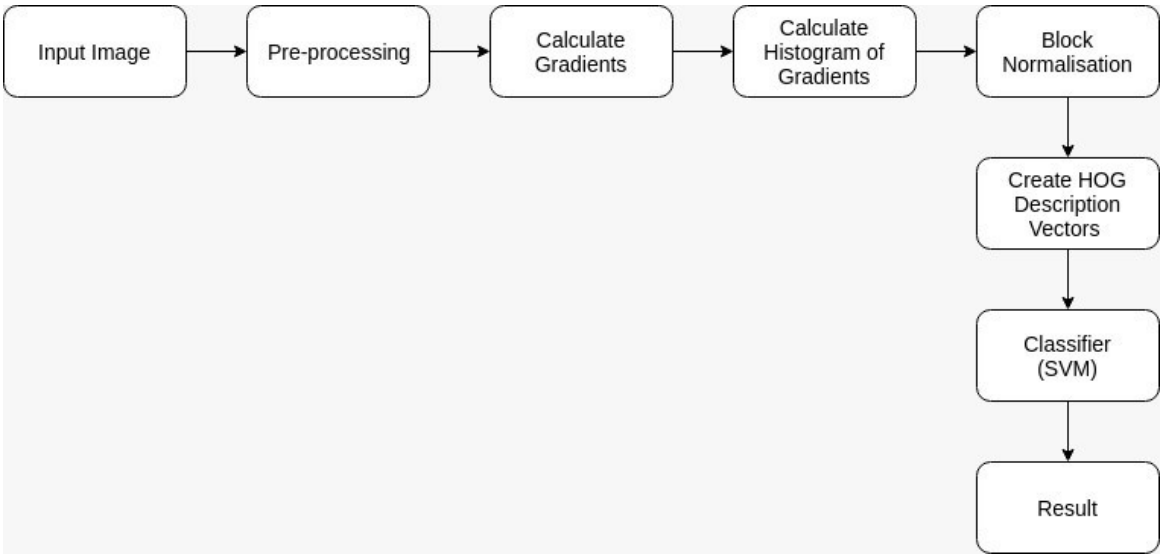


FIG 9.5 : HOG MODEL BLOCK DIAGRAM

9.1.6 Cloud Database:

A cloud database is a database service built and accessed through a cloud platform. It serves many of the same functions as a traditional database with the added flexibility of cloud computing. Users install software on a cloud infrastructure to implement the database. There are various benefits of a cloud database, some of which are the ability to pay for storage capacity and bandwidth on a per-user basis, and they provide scalability on demand, along with high availability. A cloud database also gives enterprises the opportunity to support business applications in a software-as-a-service deployment.

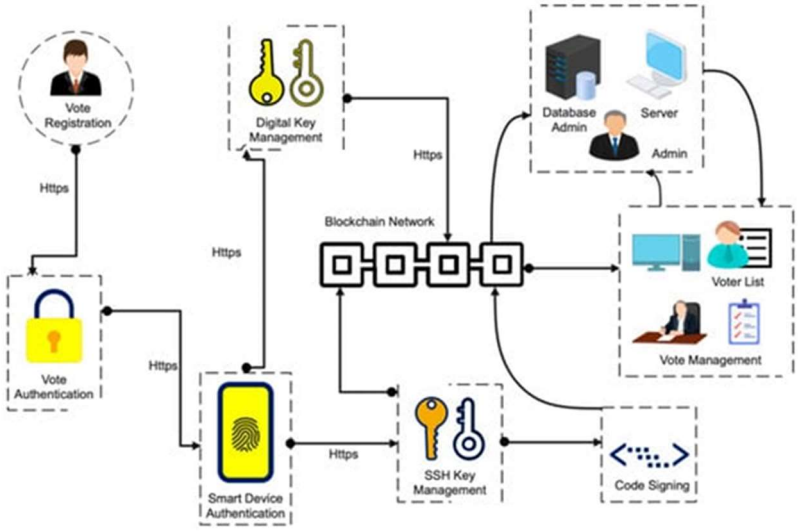


FIG 9.6 : Cloud Database

This section provides some background information on electronic voting methods. Electronic voting is a voting technique in which votes are recorded or counted using electronic equipment. Electronic voting is usually defined as voting that is supported by some electronic hardware and software. Such regularities should be competent in supporting/implementing various functions, ranging from election setup through vote storage. Kiosks at election offices, laptops, and, more recently, mobile devices are all examples of system types. Voter registration, authentication, voting, and tallying must be incorporated in the electronic voting systems

10. FUTURE SCOPE

The Research could expand in the future to include the use of an iris scanner in conjunction with a fingerprint scanner for more safe and reliable operation. It has the potential to alleviate the limits of finger print-based operations. In future we are going to propose an e voting system which is having no involvement of the human at the time voting to misuse the votes of the people. The correlation between fingerprints can be obtained in order to minimize server storage space and improve system performance. The designed unit is put to the test. We employed a face recognition technique for biometric features; however, any other biometric technique might be used. We tested the performance on 100 distinct faces and found it to be great. This could be a good way to adapt high-security voting equipment. In this paper, we have proposed a safe and secure voting system that is superior to the previous system. This technology will prevent illegitimate voters from voting and will provide faster and more accurate results. It also verifies the voter's eligibility and prohibits the same person from voting several times. It enables a person to vote from any location. We currently use finger prints, but we may be able to use iris technology in the future.

11. CONCLUSION

We conclude that by using fingerprint and facial recognition authentication for secure voting, for a more effective and transparent election process, finger print-based authentication and face recognition-based data exchange are used to address security concerns in electronic voting machines. The proposed strategy achieves better results and gives candidates more opportunities to vote in any place. It reduces human error and difficulties throughout the voting count procedure. The information must be encrypted and saved to increase the system's security. This helps to prevent data leaks while also providing better privacy to users. The votes are collected and will transfer to the head office of the voting stations in a secured manner. This avoids votes leakage. This proposed system eliminates the possibility of bogus voting.

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